

- 5 There are more than 100 species of trees and shrubs around the world with the common name of 'ironwood'.

One type of ironwood tree, *Tabebuia avellanedae*, is native to South America.

Chemicals from these trees have antibacterial properties.

- (a) (i) State the genus to which this plant belongs.

(1)

- (ii) Which structure in the cells of this plant stores starch?

(1)

- ☐ A amyloplast
☐ B middle lamella
☐ C plasmodesmata
☐ D tonoplast

- (b) Plague is a disease caused by *Yersinia pestis* bacteria.

Every year, thousands of people around the world are infected with these bacteria.

These bacteria have a thick slime capsule and many pili.

- (i) State the function of each of these structures.

(2)

Slime capsule

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Pili

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- *(ii) The ironwood tree has wood containing chemical compounds called naphthoquinones. These compounds have antibacterial properties.

Scientists are investigating whether these compounds could be used to develop a drug to treat people infected with *Y. pestis*.

Describe the methods that would be used to trial a drug containing these compounds.

(6)

(Total for Question 5 = 10 marks)



P 6 1 4 7 0 A 0 1 7 3 2